Kalman Filter

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Kalman Filter 数学推导

Optimal Recursive Data Processing Algorithm

z_k 第k次测量结果

在测量点 $z_1, z_2 \cdots z_n$

故估计值 $\hat{x}_k = \frac{1}{k}(z_1 + \cdots + z_k) = \frac{1}{k}(z_1 + z_2 + \cdots z_{k-1}) + \frac{1}{k} z_k$

Hat表 estimate $= \frac{k-1}{k}\hat{x}_{k-1} + \frac{1}{k} z_k$

$\Rightarrow$

故 $\hat{x}_k = \hat{x}_{k-1} + \frac{1}{k}(z_k - \hat{x}_{k-1})$

k个, $\frac{1}{k} \to 0$, $\hat{x}_k \to \hat{x}_{k-1}$

则 $\hat{x}_k = \hat{x}_{k-1} + K_k(z_k - \hat{x}_{k-1})$

故当前估计值 = 上一次的估计值 + 系数 x (当前测量值 − 上一次的估计值)

In KF, K_k: Kalman Gain 卡尔曼增益

估计误差 e_{EST} 测量误差 e_{MEA} $K_k = \frac{e_{EST_{k-1}}}{e_{EST_{k-1}} + e_{MEA_k}}$

~~① $e_{EST_{k-1}} \to e_A$~~

Step: ① 计算 $K_k = \frac{e_{EST_{k-1}}}{e_{EST_{k-1}} + e_{MEA_k}}$

② 计算 $\hat{x}_k = \hat{x}_{k-1} + K_k(z_k - \hat{x}_{k-1})$

③ ~~计算~~ update $e_{EST_k} = (1 - K_k) e_{EST_{k-1}}$

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Date Fusion 数据融合

~~有 z_1, z_2, z_3~~

~~有 $z_1 \sim N(\sigma_1)$~~

有 $z_1 \sim N(\mu_1, \sigma_1^2)$, $z_2 \sim ON(\mu_2, \sigma_2^2)$

估定真实值 $z = z_1 + k(z_2 - z_1)$

则 $\sigma_z^2 = (1-k)^2\sigma_1^2 + k^2\sigma_2^2$

令 $\frac{d\sigma_z^2}{dk} = 0$ 可得最优的 k

Covariance Matrix

$$Cov(X,Y) = E(XY) - E(X)E(Y)$$
$$= E((X - E(X))(Y - E(Y)))$$

$$P = \begin{bmatrix} \sigma_x^2 & \sigma_x\sigma_y & \sigma_x\sigma_z \\ \sigma_z\sigma_x & \sigma_y & \sigma_y\sigma_z \\ \sigma_z\sigma_x & \sigma_z\sigma_y & \sigma_z^2 \end{bmatrix}$$

过渡矩阵 $\begin{bmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{bmatrix} - \underbrace{\frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}}_{\text{平均值}} \begin{bmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{bmatrix}$

则 $P = \frac{1}{3}a^T a$

State Space Representation

对弹簧振动阻尼系统: ~~$m\ddot{x} = -Bx$~~

$$m\ddot{x} + B\dot{x} + kx = F = u$$

令 $x_1 = x$ $x_2 = \dot{x}$

有 $\dot{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}} = A\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + B\begin{bmatrix} u \\ 0 \end{bmatrix}$.

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~~$\begin{bmatrix} z_1 \\ z_2 \end{bmatrix}$~~ $z_1 = x_1 = x$ $z_2 = \dot{x}_1 = x_2$

$$\begin{bmatrix} z_1 \\ z_2 \end{bmatrix} = H\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

离散化有 $\begin{cases} X_k = AX_{k-1} + Bu_k + \overset{\text{过程噪声}}{w_{k-1}} \\ z_k = Hx_k + \underset{\text{测量噪声}}{v_k} \end{cases}$

problem: $\hat{x}_k$

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完整推导过程

$x_k = Ax_{k-1} + Bu_{k-1} + w_{k-1}$

$z_k = Hx_k + v_k$

$$Q = E[ww^T] = E\left[\begin{bmatrix} w_1 \\ w_2 \end{bmatrix}[w_1 w_2]\right]$$

$$= \begin{bmatrix} \sigma_{w_1} & \cdots & \sigma_{w_1 w_n} \\ \vdots & \ddots & \vdots \\ \sigma_{w_n w_1} & \cdots & \sigma_{w_n}^2 \end{bmatrix}$$

视w为高斯噪声则$E(w)=0$则$\mathrm{cov}(w_i, w_j) = E(w_i w_j)$

故可得Q

同理$R = E(vv^T)$

w_{k-1}, v_k无法建模

$\hat{x}_k^- = A\hat{x}_{k-1} + Bu_{k-1}$ 先验估计 算出来的

$\hat{x}_k^-$未加入w_{k-1}故为est值有hat，来作处理，故为先验

$\hat{x}_{k-1}$为估计而得(测量无法完全准确)

$z_k = Hx_k \rightarrow \hat{x}_{k\,MEA} = H^- z_k$ 测出来的

后验

$\hat{x}_k = \hat{x}_k^- + G(H^- z_k - \hat{x}_k^-)$ 与上文正态data fusion对应

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$G = K_k H$ 则 $K_k \in [0, H^-]$

$\hat{x}_k = \hat{x}_k^- + K_k(z_k - H\hat{x}_k^-)$

故object: find K_k 使 $\hat{x}_k \rightarrow x_k$

量化$e_k = x_k - \hat{x}_k$ 有$P = E[ee^T]$

则目标为使$\|x_k - \hat{x}_k\|$最小即$\sigma e_1^2 + \cdots \sigma e_n^2 = tr(P)$ 最小

$P = E[ee^T]$

$= E[(x_k - \hat{x}_k)(x_k - \hat{x}_k)^T]$

$x_k - \hat{x}_k = x_k - \hat{x}_k^- - K_k H x_k - K_k v_k + K_k H \hat{x}_k^-$

$= (I - K_k H)(x_k - \hat{x}_k^-) - K_k v_k$

$= (I - K_k H)\underbrace{(x_k - \hat{x}_k^-)}_{e_k^-} - K_k v_k$

由于$E(e_k^- v_k^T)$中

e_k^-与v_k无关故

$E(e_k^-)E(v_k) = 0$ 先验误差e_k^-的协方差矩阵

故$P_k = (I - K_k H) K_k P_k^- (I - K_k H)^T + K_k R K_k^T$

由$tr(A) = tr(A^T)$

得：$tr(P_k) = tr(P_k^-) - 2tr(K_k H P_k^-) + tr(K_k H P_k^- H^T K_k) + tr(K_k R K_k^T)$

则求$\frac{d\,tr(P_k)}{d K_k} = 0$

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$\frac{d\,tr(AB)}{dA} = B^T$

故 $\frac{d\,tr(P_k)}{dK_k} = -2HP_k^-{}^T + 2K_kHP_k^-H^T + 2K_kR = 0$

则 $K_k = \frac{P_k^-H^T}{HP_k^-H^T+R}$

先验估计：$\hat{x}_k^- = A\hat{x}_{k-1} + Bu_{k-1}$

后验估计：$\hat{x}_k = \hat{x}_k^- + K_k(z_k - H\hat{x}_k^-)$

卡尔曼增益：$K_k = \frac{P_k^-H^T}{HP_k^-H^T+R}$

上述唯一未知量为 P_k^-，下面推导

$P_k^- = E[e_k^-e_k^{-T}]$ $\quad e_k^- = x_k - \hat{x}_k^-$

$= A(x_{k-1} - \hat{x}_{k-1}) + w_{k-1}$

$= Ae_{k-1} + w_{k-1}$

由 x_k 与 w_k 的相互独立可化简得

$P_k^- = AE[e_{k-1}e_{k-1}^T]A^T + E[w_{k-1}w_{k-1}^T]$

$= AP_{k-1}A^T + Q$

$P_k = (I - K_kH)P_k^-$

预测：①先验估计 ②先验误差协方差

校正：① 计算 K_k，② 后验估计 ③ 更新 P_k

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EKF, Nonlinear

$x_k = f(x_{k-1}, u_{k-1}, w_{k-1})$

$z_k = h(x_k, v_k)$

$f(x_k)$ 在 $\hat{x}_{k-1}$ 处进行线性化 $\hat{x}_k$

$x_k = f(\hat{x}_{k-1}, u_{k-1}, w_{k-1}) + A(x_k - \hat{x}_{k-1}) + Ww_{k-1}$

$A = \frac{\partial f}{\partial x}\Big|_{\hat{x}_{k-1}, u_{k-1}}$, $\quad W = \frac{\partial f}{\partial w}\Big|_{\hat{x}_{k-1}, u_{k-1}}$

z_k 在 $\hat{x}_k$ 处线性化

$z_k = h(\hat{x}_k, v_k) + H(x_k - \hat{x}_k) + Vv_k$

$H = \frac{\partial h}{\partial x}\Big|_{\hat{x}_k}$ $\quad V = \frac{\partial h}{\partial v}\Big|_{\hat{x}_k}$

$P(w) \sim N(0, Q)$ $\quad P(Ww_k) \sim N(0, WQW^T)$

$P(Vv_k) \sim N(0, VRV^T)$

预测：$\hat{x}_k^- = f(\hat{x}_{k-1}, u_{k-1}, 0)$

$P_k^- = AP_{k-1}A^T + WQW^T$

校正：K_k

$\hat{x}_k = \hat{x}_k^- + K_k(z_k - h(\hat{x}_k^-, 0))$ 变化处

$P_k = (I - K_kH)P_k^-$